The GPD and compound-Poisson model

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Summary

The distribution of exceedances above a high threshold, and how to turn it into a return level.

- The GPD has two parameters: a scale $\sigma > 0$ and a shape ξ. The shape ξ is the same parameter as in the GEV.
- If block maxima follow a GEV, exceedances above a high threshold follow a GPD (Pickands-Balkema-de Haan).
- Three key assumptions: the exceedances are independent, the threshold is high enough for the GPD to hold, and the process is stationary.
- The compound-Poisson model combines a Poisson arrival rate λ (exceedances per year) with the GPD for exceedance sizes to give annual exceedance probabilities.
- The return level at return period T is $x_T = u + \frac{\sigma}{\xi}\left[(T\lambda)^{\xi} - 1\right]$.

Key results.

$$G(d) = 1 - \left(1 + \frac{\xi d}{\sigma}\right)^{-1/\xi}, \qquad d > 0$$

$$P(\text{annual max} > x) = 1 - \exp\big(-\lambda\,[1 - G(x - u)]\big)$$

$$x_T = u + \frac{\sigma}{\xi}\left[(T\lambda)^{\xi} - 1\right]$$

Bibliography